

MSCI 217 – Groupwork assignment

Business Forecasting in Practice

During an internship at a large online company, you are asked to analyse historical sales data to improve their current forecasting practices.

Prepare a **company report** outlining your analysis and findings.

The sales data for product P1 and P2 is weekly and data was collected until November 12, 2023. Products P3 and P4 are monthly and might be related. For product P4, the company has also provided the average price that was set for this product per month. The last month for which sales data for products P3 and P4 was collected is August 2023.

They have asked you to complete the following tasks:

- 1) Use data exploration and time-series decomposition to identify potential structures in the data.
 - Document and describe your results.
 - Visualize the data using different plots (e.g., time series, seasonal) and describe your observations in your own words.

[20% of grade]
- 2) After conducting the data exploration and time-series decomposition in task 1), formulate at least three questions that you would ask the company.
 - Write down the three questions in your report.
 - Explain the rationale why you would like to ask these questions and how their answers would help you with your analysis.

[10% of grade]
- 3) What are sensible candidate methods to forecast each product? How would the answers in question 2 help you with your choice? Justify the methods you propose and explain the reasons which lead to this choice. Consider at least one time-series and one causal method where appropriate. Discuss strengths and weaknesses of the methods considered.

[20% of grade]
- 4) Split your data in in-sample and out-of-sample dataset. Define an appropriate out-of-sample dataset and comment why it is appropriate for your data. Describe which error measures you would choose for the methods selected in 3) and explain why. Do not select error measures arbitrarily but justify your choice.

[10% of grade]
- 5) Apply the methods proposed in 3) and use the error measures from 4) to determine which forecasts the company should choose. Provide evidence that the forecasting methods have been appropriately applied to each time series (parameter details, etc). Use 1- and 5-period-step ahead forecasts for your analysis. Which method performs best? Do you still agree with your previous proposals or do you need to make any adjustments. Explain why you think your choice is justified or why you need to make adjustments. How do the number of steps of your forecast horizon affect the forecasts and why? Explain and visualize your results.

[20% of grade]
- 6) Interpret and discuss your results. Which method(s) would you recommend to the company? Critically discuss the robustness of your results as well as potential reasons for performance differences. Provide a conclusion how the company should proceed and what they should learn from your analysis.

[10% of grade]

Your company report should consist of no more than **12 pages** (only main text). **Reports of excessive length will be penalised.** Cover sheet, table of contents and appendix do not count towards the page limit. Make adequate use of graphs to support your arguments and tables to compare results. For formatting, use single spacing, format normal text in Calibri, font size 11, text in tables, figure and table headings in font size 10, and leave 2cm of margin left and right. Make sure that your names, student ids and the number of your dataset are stated on the cover sheet/first page of your report.

You need to be critical which graphs are necessary in the report and do not simply add everything else to the appendix. Be critical which information is needed to communicate the validity and completeness of your analysis, as well as your findings. Ensure that the main report is readable without consulting the appendix. The appendix should only contain additional details (which are referred to from the main text).

10% of the grade will be evaluated based on clarity and structure of the company report.

The report should be homogeneous, using the same rigor in producing and evaluating the predictions. The report should be written as if for a specialist in OR (incl. executive summary, introduction and conclusions / summary, numbered headings etc.) and should allow an understanding of your choices and replication of your experiments.

This is a **group** assignment. You will receive one dataset per group. The assignment is due on **May 7 at 14:00**. Standard departmental penalties will apply for late work unless you have been given an extension for exceptional reasons from the course administrator. All submissions will be checked by plagiarism software. You must submit your report, R-scripts and data-files **electronically through Moodle**.

Good Luck!

*Please use **Moodle** for being assigned to a group or for creating your own group.
You will receive a dataset when the groups are set up.*